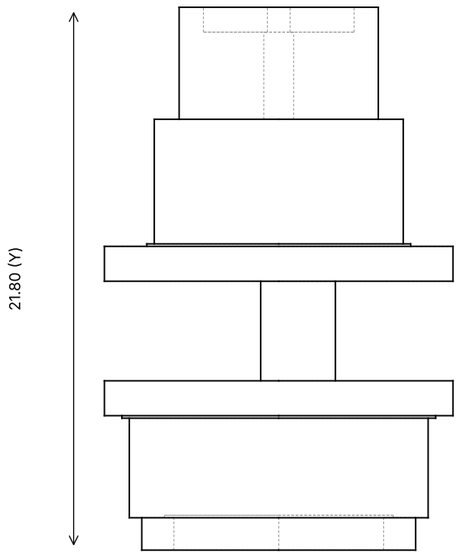
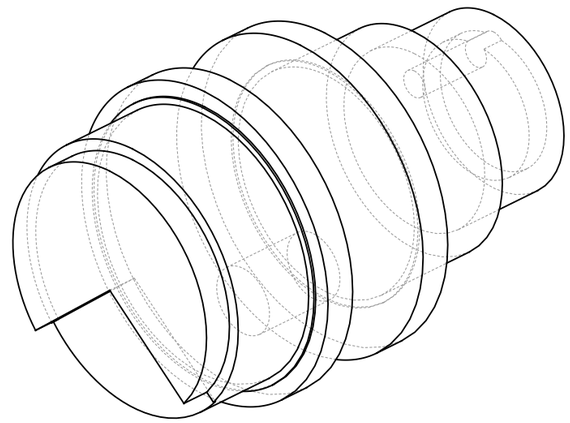
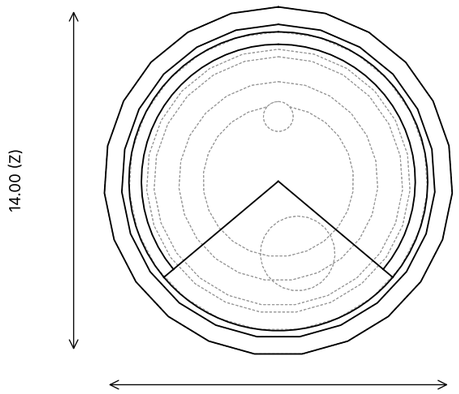
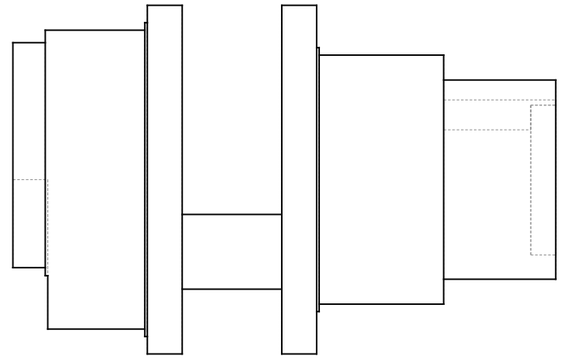




TOP (along -Z)



ISO (iso, not to scale)



ENG-PD-04   Pintle crankshaft (throw r 3.0, pin 3, journals 12 / 10, dog end, spring hub, magnet pocket)
Material: 17-4PH (1.4542) condition H900: pintle crankshaft, crank/link pins, MV-P poppet   Process: CNC lathe + mill (eccentric pin turned in a second setup) + grind journals, H900
Mass: 12.5 g (CAD volume x density)   Units: mm   Frame: modelling frame (+Z forward)   Bounding box: 14.00 x 21.80 x 14.00
Datum: journal A axis A   Finish: ground journals Ra 0.4; passivate   Heat treatment: 17-4PH H900 (AMS 5643)
Fits / features: journal A 12 k5; journal B 10 k5 coaxial 0.005 to A; pin 3 g6, offset 3.000 +/-0.01, parallel to A 0.01; spring hub 8 h9
Notes: closed at phi 15 deg (shown), mechanical stop 113.09 deg = 4.2 mm stroke; pin span 4 mm between the webs (pintle_v2p4 jam margins use a 6 mm simply-supported span: conservative); dog notch 100 deg for the PD-05 tooth (on
ARGOS engine CAD v3.5 - DESIGN-MODEL DRAWING generated from the solid; dimensions from the model; general tolerances ISO 2768-m unless a fit is listed; NOT released for fabrication (Mechanical, 2026-10-13)